

KONGU ENGINEERING COLLEGE

AIR POLLUTION MONITORING SYSTEM USING MQ2 GAS SENSOR

EXCERSICE - 11

HEMAVARSHINI S – 24MER017

KAVIN KUMARAN S – 24MER023

KOWSIKA K – 24MER026

VISHNUPRASATH B – 24MER062



KONGU ENGINEERING COLLEGE

— TRANSFORM YOURSELF —

AIR POLLUTION MONITORING SYSTEM USING MQ2 GAS SENSOR

1. PROJECT OVERVIEW

The Air Pollution Monitoring System using MQ2 Gas Sensor is a safety and environmental monitoring system designed to detect the presence of harmful and combustible gases in the surrounding atmosphere.

The system uses an MQ2 gas sensor to sense gases such as smoke, LPG, methane and other combustible gases. The analog output from the MQ2 sensor is processed by an Arduino UNO. Based on the detected gas level, the system displays the air quality status on a 16×2 LCD and activates a buzzer and warning LED when the gas concentration exceeds the predefined threshold.

This system can be used in:

- Industrial environments
- Laboratories
- Workshops
- Kitchens
- Parking areas
- Indoor air-quality monitoring
- Environmental safety applications

Main Objectives

1. Detect harmful or combustible gases.
2. Monitor the gas level continuously.
3. Provide an immediate warning when the gas level becomes high.
4. Display the air-quality condition to the user.
5. Develop a low-cost and simple pollution monitoring system.
6. Simulate and test the system before hardware implementation.

2. PROBLEM STATEMENT

Air pollution caused by harmful gases and smoke can create serious health and safety problems. In industrial and domestic environments, gas leakage or excessive smoke may not be detected immediately by humans. Conventional monitoring systems can be expensive and complicated for small-scale applications. Therefore, there is a need for a low-cost, real-time air pollution monitoring system that can detect harmful gases and provide an immediate warning to users. The proposed system uses an MQ2 gas sensor and Arduino UNO to continuously monitor the surrounding air and alert the user when the gas level exceeds a predefined threshold.

3. PROPOSED SOLUTION

The proposed system consists of an MQ2 gas sensor, Arduino UNO, LCD display, LED indicators and buzzer.

The MQ2 sensor detects the presence of smoke and combustible gases. Its analog output changes according to the gas concentration. The Arduino reads this analog value and compares it with predefined threshold values.

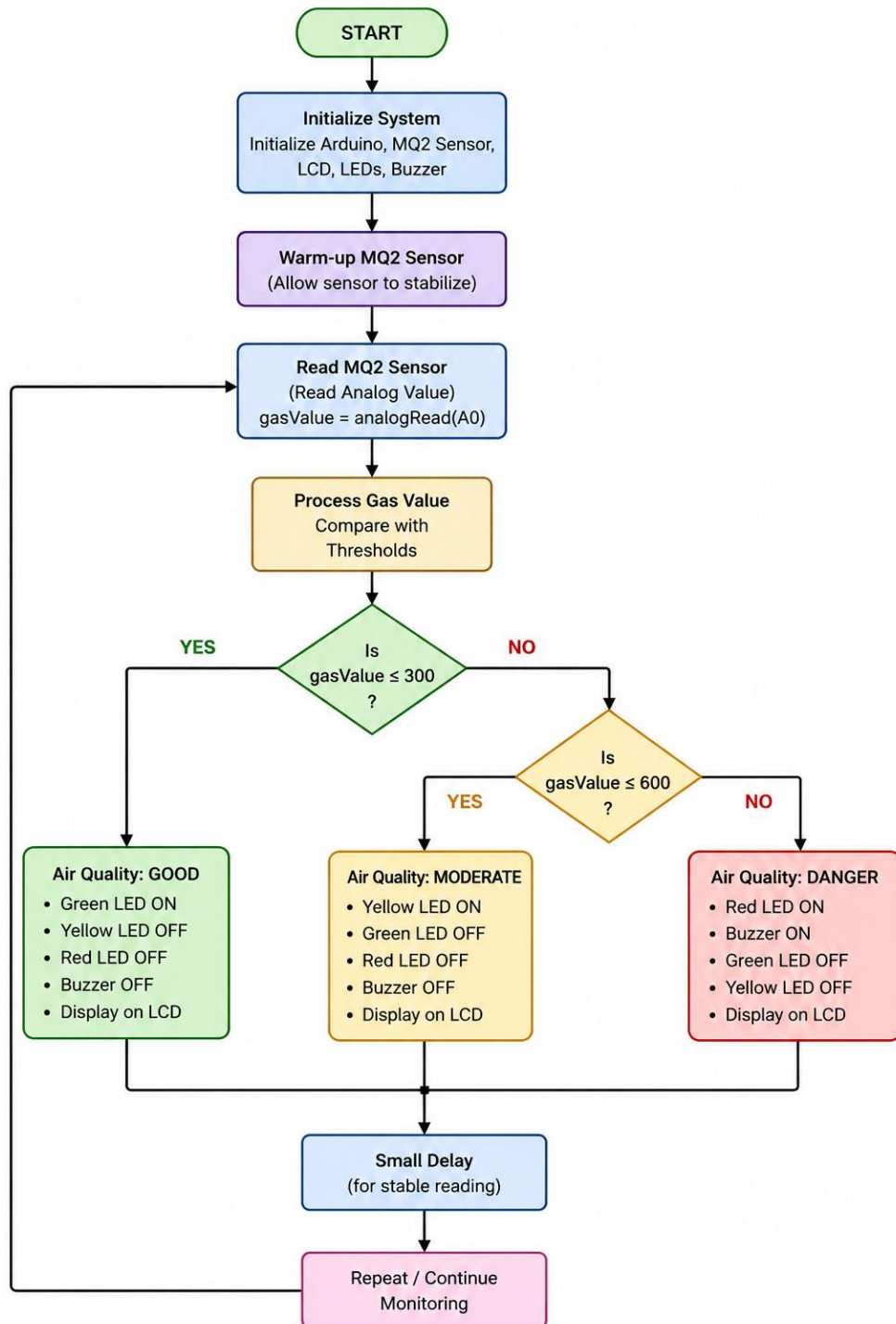
Working Principle

Air → MQ2 Sensor → Arduino UNO → Gas Level Analysis → LCD + LED + Buzzer

The system operates in three basic conditions:

Gas Sensor Value	Condition	Output
Low	Good Air Quality	Green LED ON
Medium	Moderate Pollution	Yellow LED ON
High	Dangerous Gas Level	Red LED + Buzzer ON

4. SYSTEM ARCHITECTURE



5. CIRCUIT TABLE

S.No.	Component	Component Pin	Arduino UNO Pin / Connection
1	MQ2 Gas Sensor	VCC	5V
2		GND	GND
3		AO	A0
4		DO	Not Used
5	Green LED	Anode (+)	D8 through 220 Ω resistor
6		Cathode (-)	GND
7	Yellow LED	Anode (+)	D9 through 220 Ω resistor
8		Cathode (-)	GND
9	Red LED	Anode (+)	D10 through 220 Ω resistor
10		Cathode (-)	GND
11	Buzzer	Positive (+)	D11
12		Negative (-)	GND
13	16×2 LCD	VSS	GND
14		VDD	5V
15		V0	Potentiometer middle pin
16		RS	D2
17		RW	GND
18		E	D3
19		D4	D4
20		D5	D5
21		D6	D6

22		D7	D7
23		A (LED+)	5V
24		K (LED-)	GND
25	Potentiometer	One side	5V
26		Middle pin	LCD V0
27		Other side	GND

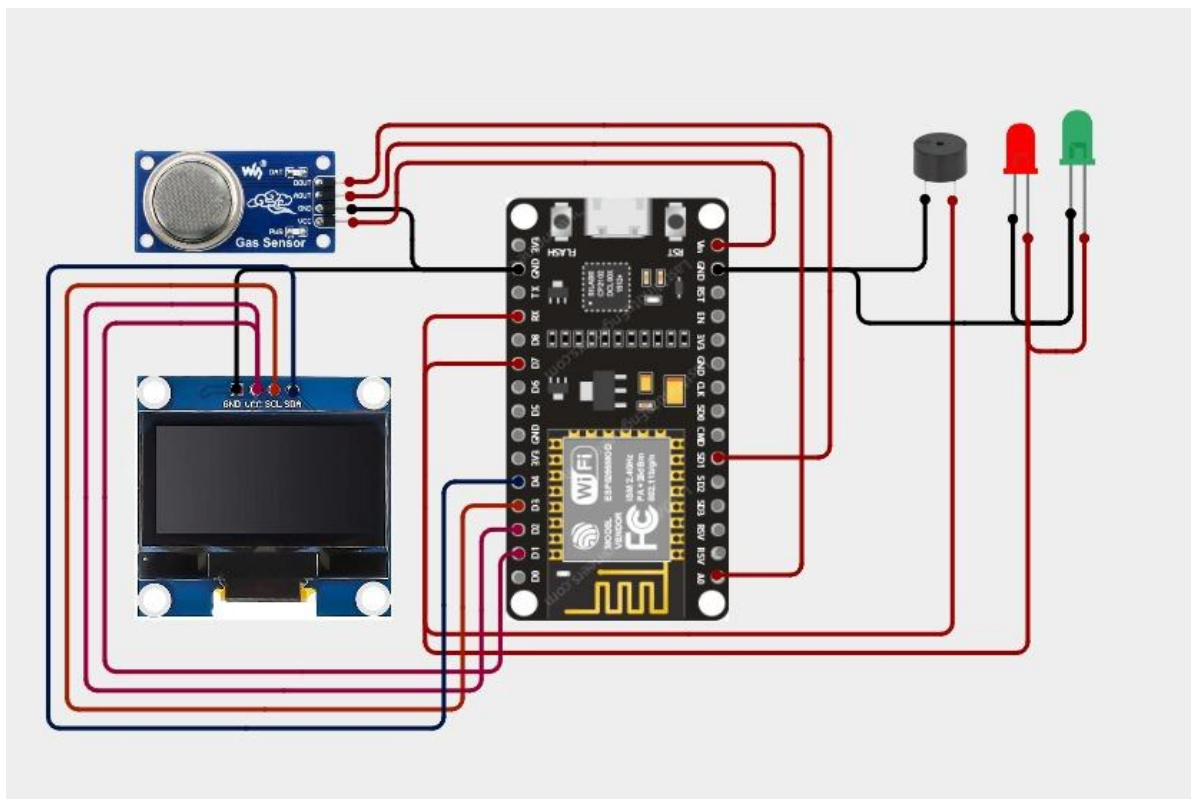
6. CIRCUIT DESIGN

The MQ2 sensor is connected to the analog input **A0** of the Arduino UNO.

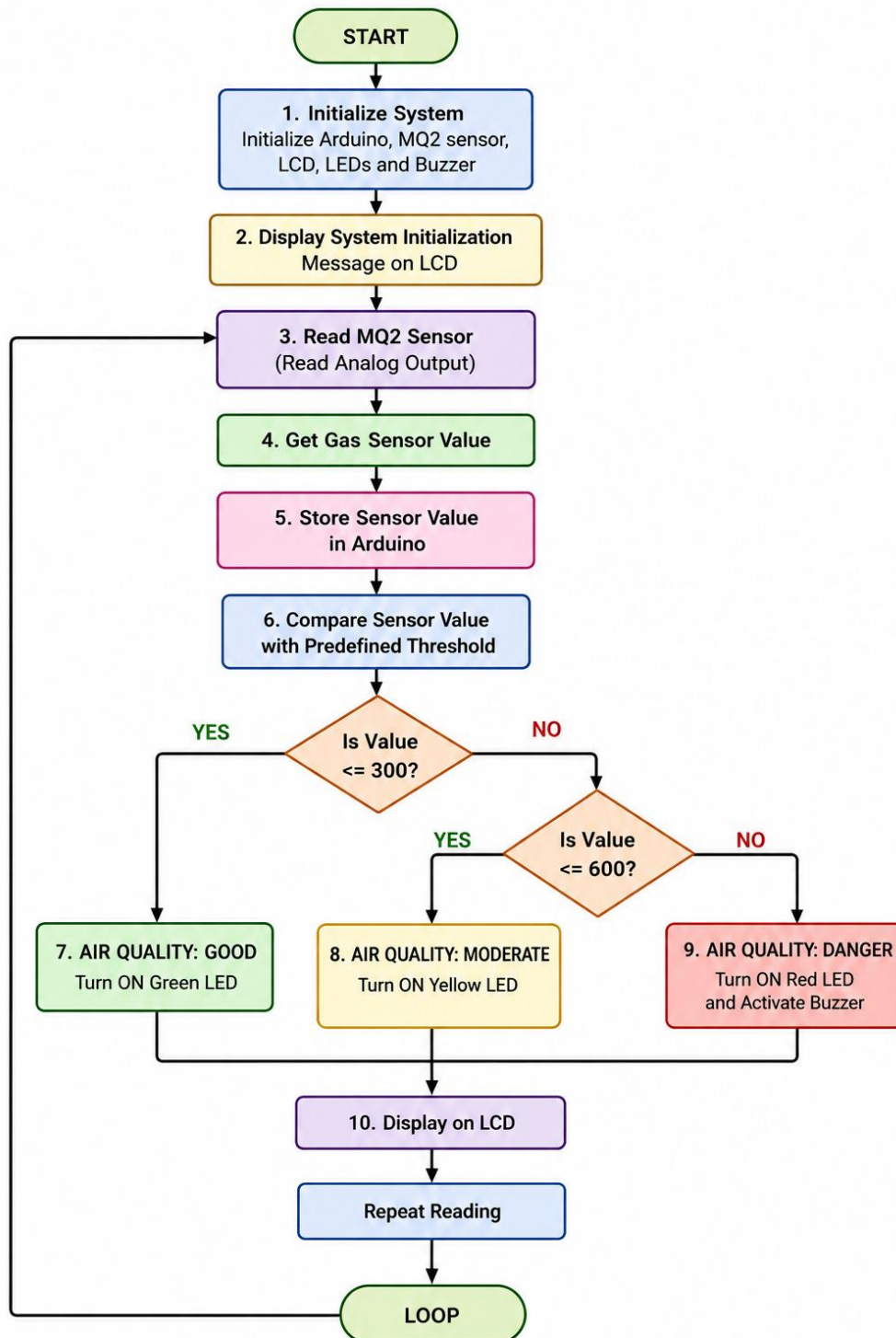
The sensor is powered using the 5V supply from the Arduino. The analog output of the MQ2 sensor changes according to the detected gas concentration.

The Arduino continuously reads this analog value using its built-in ADC.

The output devices are connected as follows:



7. ALGORITHM



8. CODING

```
#include <LiquidCrystal.h>
```

```
// -----
```

```
// LCD CONNECTIONS
```

```
// RS = D2
```

```
// E = D3
```

```
// D4 = D4
```

```
// D5 = D5
```

```
// D6 = D6
```

```
// D7 = D7
```

```
// -----
```

```
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
```

```
// MQ2 Sensor
```

```
const int MQ2_PIN = A0;
```

```
// LEDs
```

```
const int GREEN_LED = 8;
```

```
const int YELLOW_LED = 9;
```

```
const int RED_LED = 10;
```

```
// Buzzer
```

```
const int BUZZER = 11;
```

```
// Threshold values
```

```
const int GOOD_LIMIT = 300;
```



```
const int MODERATE_LIMIT = 600;

void setup()
{
    // Set LED and buzzer pins as OUTPUT
    pinMode(GREEN_LED, OUTPUT);
    pinMode(YELLOW_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);

    // Start LCD
    lcd.begin(16, 2);

    // Start Serial Monitor
    Serial.begin(9600);

    // Turn OFF all indicators initially
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(YELLOW_LED, LOW);
    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER, LOW);

    // -----
    // SYSTEM INITIALIZATION MESSAGE
    // -----

    lcd.clear();
```

```
lcd.setCursor(0, 0);  
lcd.print("Air Pollution");
```

```
lcd.setCursor(0, 1);  
lcd.print("Monitoring...");
```

```
delay(2000);
```

```
// -----  
// MQ2 WARM-UP MESSAGE  
// -----
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("MQ2 Sensor");  
lcd.setCursor(0, 1);  
lcd.print("Warming Up...");
```

```
delay(5000);
```

```
lcd.clear();  
}
```

```
void loop()  
{  
  // -----  
  // STEP 1: READ MQ2 SENSOR
```

```
// -----  
int gasValue = analogRead(MQ2_PIN);  
  
// Display sensor value in Serial Monitor  
Serial.print("MQ2 Gas Value: ");  
Serial.println(gasValue);  
  
// -----  
// TURN OFF ALL OUTPUTS BEFORE NEW DECISION  
// -----  
digitalWrite(GREEN_LED, LOW);  
digitalWrite(YELLOW_LED, LOW);  
digitalWrite(RED_LED, LOW);  
digitalWrite(BUZZER, LOW);  
  
// -----  
// STEP 2: CHECK GAS LEVEL  
// -----  
  
// GOOD AIR QUALITY  
if (gasValue <= GOOD_LIMIT)  
{  
    digitalWrite(GREEN_LED, HIGH);  
  
    lcd.clear();
```

```
lcd.setCursor(0, 0);  
lcd.print("Gas: ");  
lcd.print(gasValue);  
  
lcd.setCursor(0, 1);  
lcd.print("AIR: GOOD");  
  
Serial.println("AIR QUALITY: GOOD");  
}  
  
// MODERATE AIR QUALITY  
else if (gasValue <= MODERATE_LIMIT)  
{  
    digitalWrite(YELLOW_LED, HIGH);  
  
    lcd.clear();  
  
    lcd.setCursor(0, 0);  
    lcd.print("Gas: ");  
    lcd.print(gasValue);  
  
    lcd.setCursor(0, 1);  
    lcd.print("AIR: MODERATE");  
  
    Serial.println("AIR QUALITY: MODERATE");
```

```
}
```

```
// DANGEROUS AIR QUALITY
```

```
else
```

```
{
```

```
    digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(BUZZER, HIGH);
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("Gas: ");
```

```
    lcd.print(gasValue);
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("AIR: DANGER");
```

```
    Serial.println("AIR QUALITY: DANGER");
```

```
}
```

```
// -----
```

```
// STEP 3: WAIT BEFORE NEXT READING
```

```
// -----
```

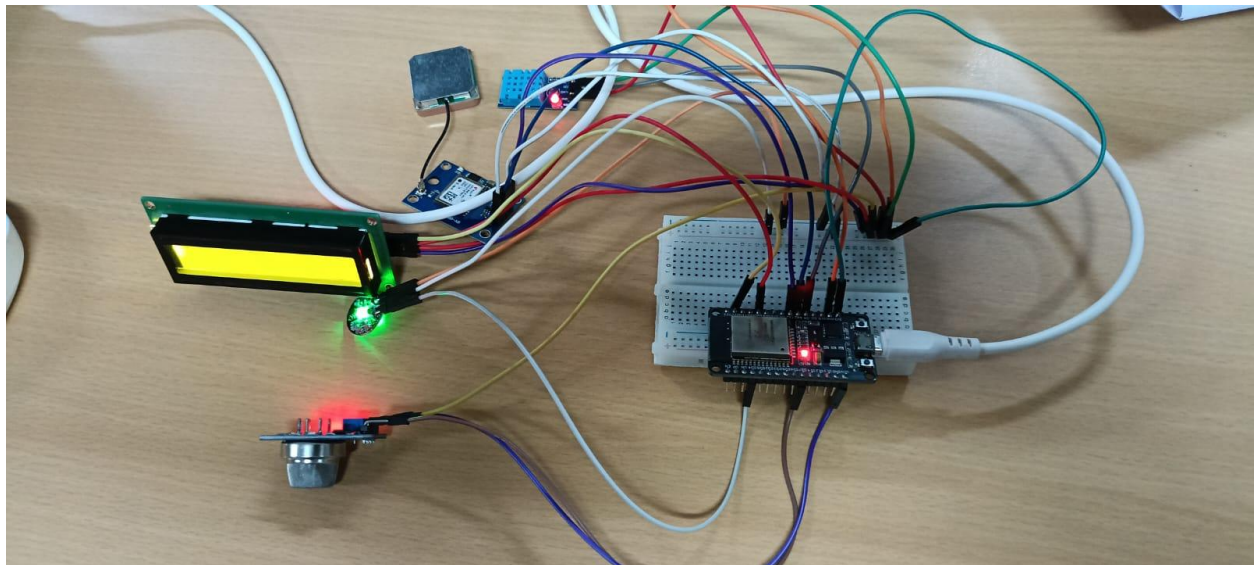
```
delay(1000);
```

```
}
```

9. SYSTEM WORKING

The MQ2 gas sensor is the primary sensing element in the system. When the system is powered ON, the Arduino initializes all connected components. The MQ2 sensor requires a warm-up period for stable operation. After initialization, the Arduino reads the analog output from the MQ2 sensor through pin A0.

The sensor value is continuously compared with the predefined threshold values.



10. BILL OF MATERIALS

S.No	Component	Quantity
1	Arduino UNO	1
2	MQ2 Gas Sensor	1
3	16×2 LCD	1
4	Green LED	1
5	Yellow LED	1
6	Red LED	1
7	Buzzer	1

8	220 Ω Resistor	3
9	10 k Ω Potentiometer	1
10	Breadboard	1
11	Jumper Wires	As required
12	USB Cable	1
13	5V Power Supply	1

11. PROTOTYPE IMPLEMENTATION

The prototype can be implemented using an Arduino UNO connected to the MQ2 gas sensor and output devices.

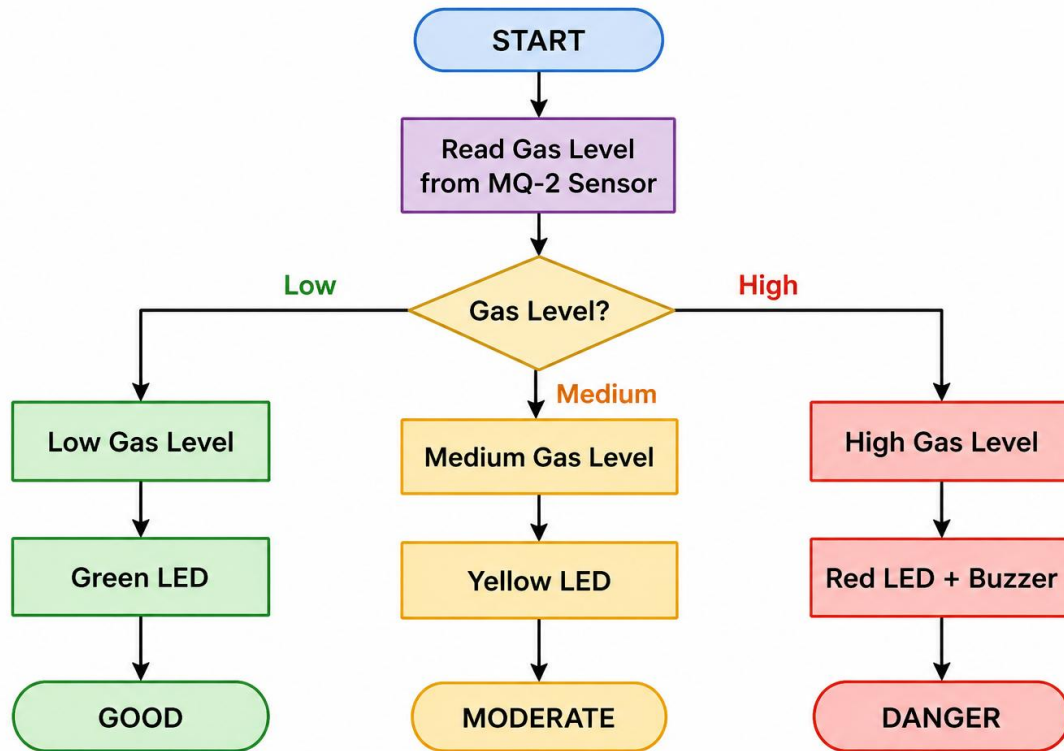
Prototype Procedure

1. Connect the MQ2 sensor to the Arduino.
2. Connect the LCD display.
3. Connect green, yellow and red LEDs through suitable resistors.
4. Connect the buzzer to the Arduino.
5. Upload the Arduino program.
6. Allow the MQ2 sensor to warm up.
7. Monitor the sensor readings through the LCD or Serial Monitor.
8. Introduce smoke or a suitable test gas source near the sensor in a controlled manner.
9. Observe the changes in the sensor value.
10. Verify the LED and buzzer response.

Simulation

The circuit can be simulated using an online electronics simulator such as **Wokwi**.

During simulation, the MQ2 gas level can be varied to verify the three operating conditions:



12. RESULTS & PERFORMANCE ANALYSIS

The developed system successfully demonstrates the basic concept of real-time air-pollution and gas monitoring.

Expected Results

Test Condition	Sensor Value	LED	Buzzer	LCD Status
Clean/Low Gas	0–300	Green	OFF	GOOD
Moderate Gas	301–600	Yellow	OFF	MODERATE
High Gas	601–1023	Red	ON	DANGER

Advantages

1. Low-cost system.
2. Easy to build and operate.
3. Real-time monitoring.
4. Immediate warning during high gas levels.
5. Suitable for educational and prototype applications.
6. Can be expanded with Wi-Fi or IoT functionality.

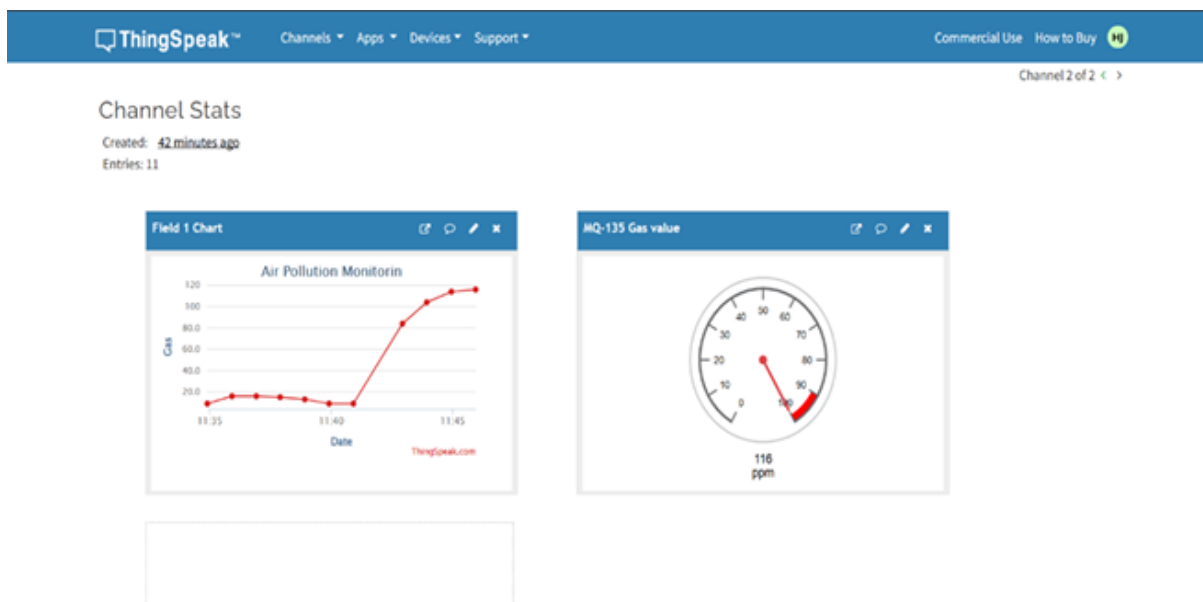
Limitations

1. MQ2 is a general-purpose gas sensor and responds to multiple gases.
2. Accurate ppm measurement requires calibration.
3. Sensor readings can be affected by temperature and humidity.
4. The sensor requires a warm-up period.
5. The threshold values used in simulation are not universal safety limits.

Future Scope

The system can be improved by adding:

- ESP32/ESP8266 for IoT connectivity.
- Mobile application monitoring.
- Cloud-based data storage.
- Real-time graphical monitoring.
- Multiple gas sensors.
- Automatic ventilation/fan control.
- GPS location tracking for outdoor monitoring.
- Data logging and historical analysis.



CONCLUSION

The Air Pollution Monitoring System using MQ2 Gas Sensor provides a simple and economical solution for detecting harmful gases and monitoring air-quality conditions. The MQ2 sensor detects changes in gas concentration and sends an analog signal to the Arduino UNO. The Arduino processes the sensor value and provides appropriate visual and audible indications through the LCD, LEDs and buzzer.

The proposed system can be used as a basic safety-monitoring system in industrial, domestic and environmental applications. With the addition of IoT technology, the system can be further developed into a remote and real-time air-quality monitoring platform.